ECE 501 DIGITAL IMAGE PROCESSING
SAR Optical Feature Fusion For Crop Type Classification
REPORT WEEK 7
GROUP 11

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1) Introduction And Objectives

Google Earth Engine was used to access the Sentinel-1 SAR data in google earth engine with the dataset name COPERNICUS/S1_GRD.

The extraction workflow was to concentrate on:

Choosing Sentinel-1 GRD data.

- IW (Interferometric Wide) acquisition mode filtering.
- Choosing VV and VH polarization bands.
- Keeping the direction of the orbit the same.

Applying date filters to the date that the YieldSAT field was acquired.

Creating median composites for the specified date ranges.

Using “reduceRegion()” at 10 metre spatial resolution to extract mean VV and VH backscatter values.

Geometries in the field were a key component to the extraction process. To assure the extraction of SAR statistics from the appropriate agricultural areas, a validation and geometry construction process was needed.

SAR Feature Analysis

- The following were the main SARs that were taken into consideration:
- VV: Vertically transmitted and received polarization radar.
- VH: Cross polarized backscattered data showing information on structure of vegetation.
- DVS (VV/VH Ratio): Another parameter to assess the differences between co-polarized and cross-polarized responses.
- The features are useful to represent the structure and surface characteristics of the vegetation that can not be fully captured from optical imagery.

2) Model Evaluation

a) SAR

The classification model based on SAR data was compared with the Optical-only and Fusion models.

The following were used as evaluation criteria:

- Accuracy
- Precision
- Recall
- Macro-F1 Score
- Weighted-F1 Score

SAR-only results aid in assessing the ability of radar information to separate the various crop types, especially corn from soybean.

Model	Accuracy	Precision	Recall	Macro-F1	Weighted-F1
Logistic Regression	0.866	0.839	0.857	0.847	0.867
Random Forest	0.870	0.859	0.824	0.835	0.863

SVM RBF	0.885	0.858	0.878	0.867	0.886
XGBoost	0.876	0.858	0.842	0.848	0.873

Among the four classifiers, SVM RBF has achieved the best SAR-only performance, with an accuracy of 0.885 and a Macro-F1 score of 0.867. This shows that SAR features provide useful information for identifying the three crop types.

b) Optical

The Optical-only classification models were evaluated using Sentinel-2 features to understand how well optical data can distinguish between corn, soybean and wheat.Three classification algorithms, Logistic Regression, Random Forest and SVM, were used for the evaluation. The models were assessed using 5-fold spatial cross-validation.

Model	Accuracy	Precision	Recall	Macro-F1	Weighted-F1
Logistic Regression	0.884	0.855	0.869	0.862	0.885
Random Forest	0.785	0.754	0.696	0.717	0.774
SVM RBF	0.847	0.816	0.813	0.815	0.847
XGBoost	0.835	0.822	0.769	0.790	0.829

Logistic Regression got the best performance from the Optical-only models, with accuracy of 0.884 and Macro-F1 score of 0.862. This shows that optical features provide useful information for identifying between corn, soybean and wheat.

Optical features such as NDVI, NDRE and NDWI helped capture differences in vegetation conditions and crop growth patterns. However, some confusion between crop classes was observed, particularly between soybean and wheat. This indicates that optical information alone may not always be sufficient to distinguish crops with similar vegetation characteristics.

c) Fusion

The Fusion approach combines Sentinel-1 SAR features with Sentinel-2 Optical features. The main objective is to use the complementary information provided by both satellite data sources to improve crop classification.

Model	Accuracy	Precision	Recall	Macro-F1	Weighted-F1
Logistic Regression	0.958	0.954	0.950	0.948	0.958
Random Forest	0.902	0.888	0.886	0.881	0.902
SVM RBF	0.928	0.916	0.914	0.914	0.928
XGBoost	0.920	0.904	0.903	0.903	0.920

Logistic Regression got the best performance among the Fusion models, with an accuracy of 0.958 and a Macro-F1 score of 0.948. This shows the benefit of combining SAR and Optical features for crop type classification.

The Fusion model combines vegetation-related information from Optical data with radar backscatter and crop structural information from SAR data. This allows the models to consider more characteristics of agricultural fields, resulting in better classification performance.

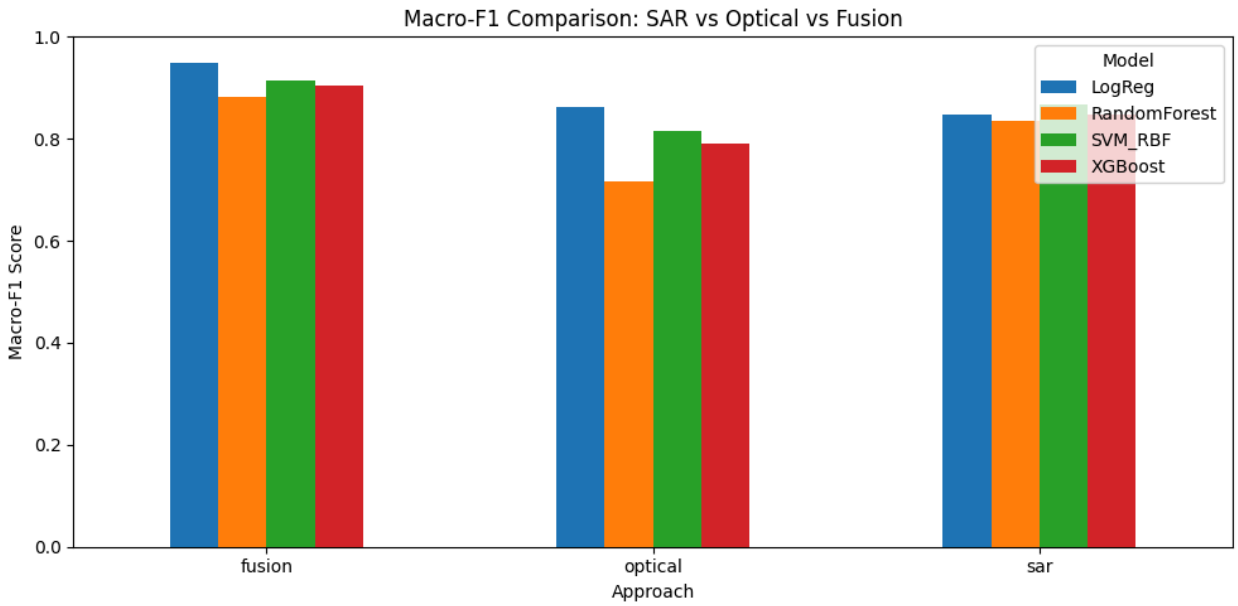
The Random Forest Fusion feature importance analysis showed that SAR features contributed 63%, while Optical features contributed 37%. This indicates that both data sources provide useful information to the classification process.

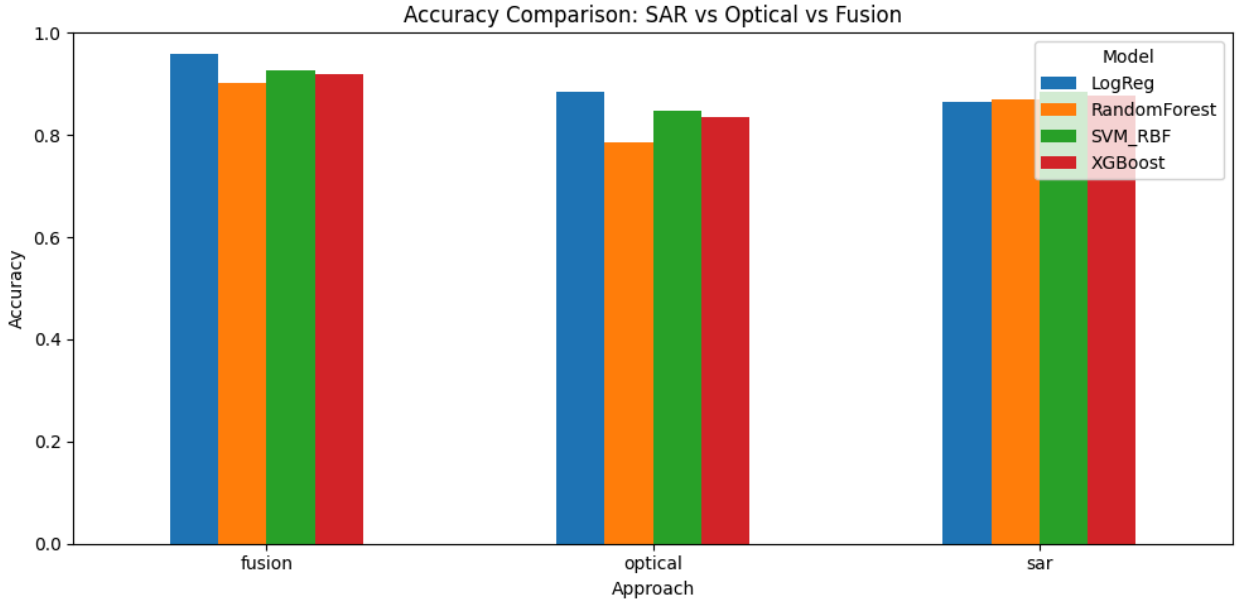
d) Comparative Analysis

The results show that the Fusion model performed better than both SAR-only and Optical-only models across all four classifiers. Logistic Regression got the highest Macro-F1 score of 0.948 with Fusion.

Model	SAR Macro-F1	Optical Macro-F1	Fusion Macro-F1
Logistic Regression	0.847	0.862	0.948
Random Forest	0.835	0.717	0.881
SVM RBF	0.867	0.815	0.914
XGBoost	0.848	0.790	0.903

This improvement is mainly because Sentinel-2 provides information about vegetation and crop growth, while Sentinel-1 gives crop structure and radar backscatter. Combining both improves the classification of corn, soybean and wheat.





e) Confusion Matrix Analysis:

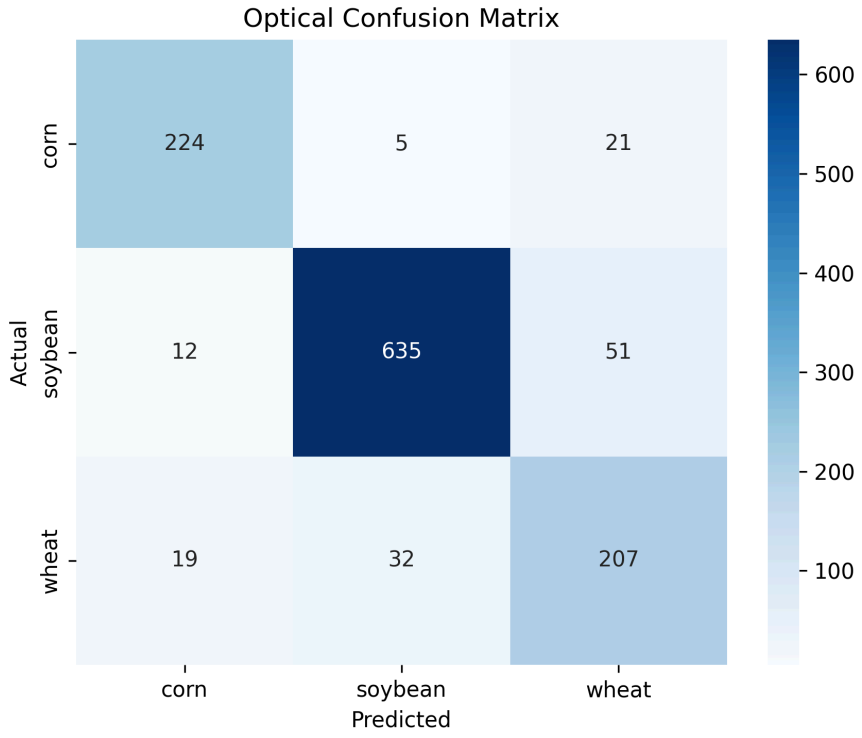
- ## Optical-Only Classification

  The confusion matrix of the Optical-only Logistic Regression model shows how accurately the three crop types were classified. The model correctly classified 224 corn fields, 635 soybean fields and 207 wheat fields.

Soybean had the highest number of correct predictions. Some confusion was observed between soybean and wheat, mainly because their vegetation characteristics can be similar.

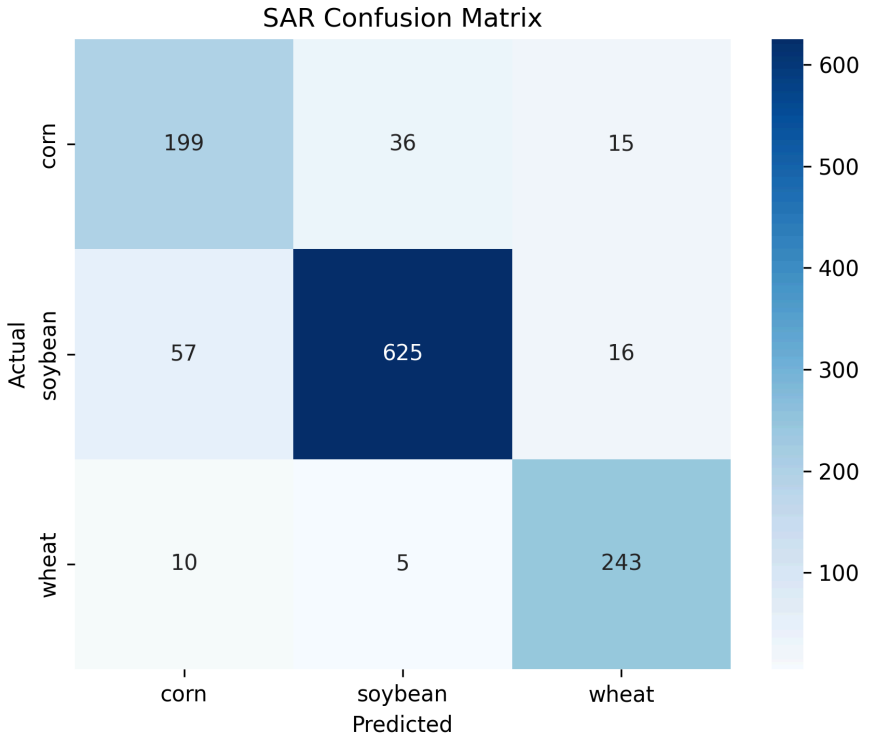
Actual / Predicted	Corn	Soybean	Wheat
Corn	224	5	21
Soybean	12	635	51
Wheat	19	32	207

Overall, optical features were useful for crop classification, although some misclassifications were still present.



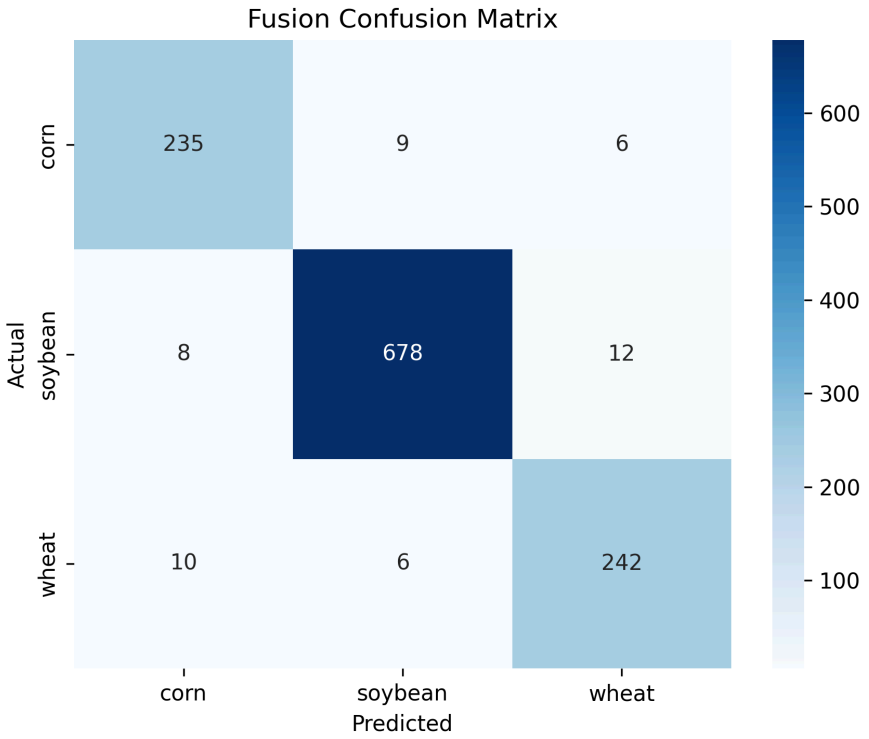
- **SAR - SVM RBF**

Actual / Predicted	Corn	Soybean	Wheat
Corn	199	36	15
Soybean	57	625	16
Wheat	10	5	243



- **Fusion - Logistic Regression**

Actual / Predicted	Corn	Soybean	Wheat
Corn	235	9	6
Soybean	8	678	12
Wheat	10	6	242



3) Feature Importance Analysis

- For understanding the contribution of the radar information to the crop classification, SAR feature importance was analysed.
- The main features taken into consideration were VV, VH and VV/VH. They were compared with optical characteristics like NDVI, NDRE and NDWI in terms of their relative importance.
- This comparison is used to find out if SAR adds information to the optical vegetation indices.

Crop	Mean NDVI	Peak NDVI	Mean NDRE	Mean NDWI
Corn	0.484	0.779	0.398	-0.446
Soybean	0.515	0.822	0.405	-0.483
Wheat	0.491	0.756	0.391	-0.452

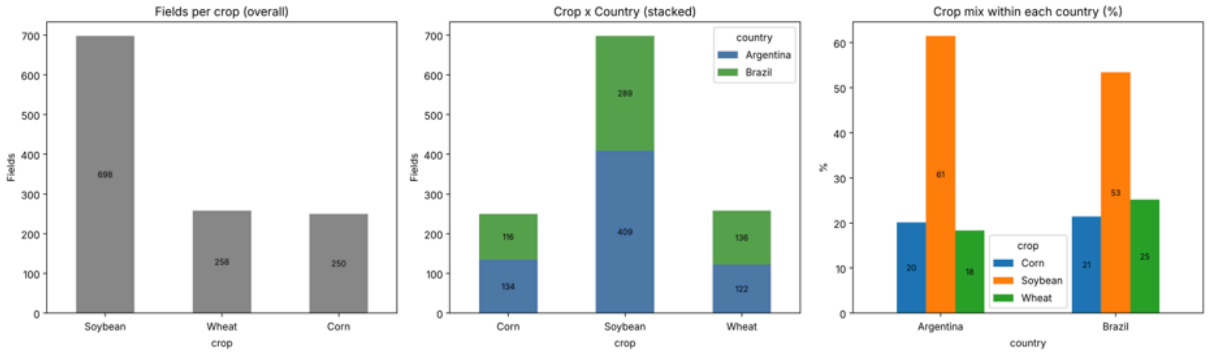
SAR - Optical Feature Contribution

- SAR features contributed 63% to the model, but optical features also played an important role and contributed upto 37% to the model. This shows that combining both types of satellite data provides useful information for better crop classification.

4) Country-wise And Dataset

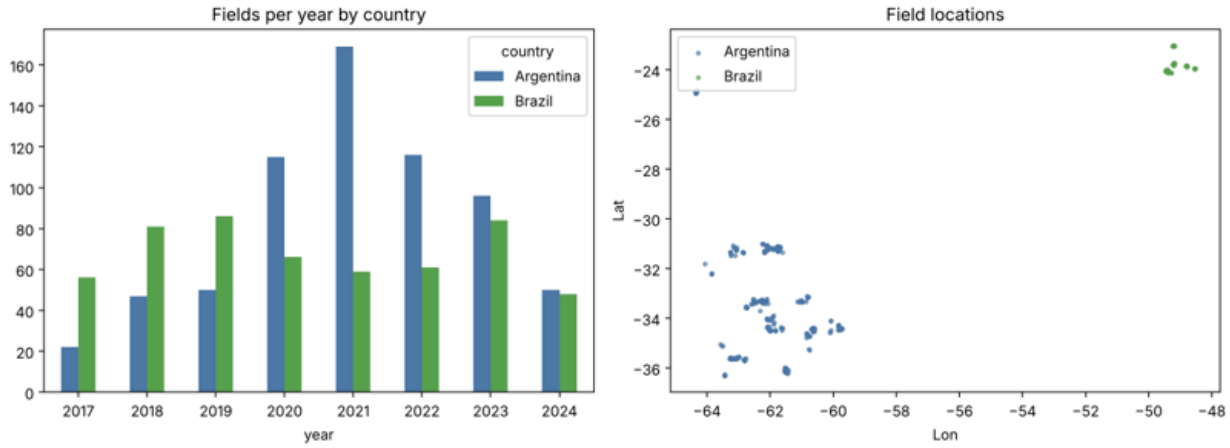
Item	Value
Total Fields	1206
Countries	2(Argentina, Brazil)
Crops	3(Soyabean, corn, Wheat)
Years	2017-2024
Spatial blocks	76
Mean Season Length	159 days

Crop-wise Distribution



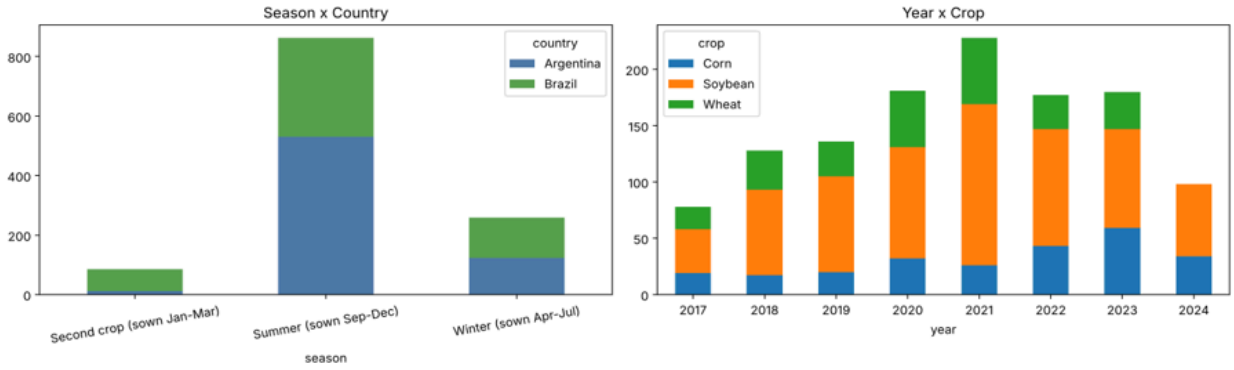
- Soybean was the predominant crop in both acres (698 fields, or 57.9%) and number of fields (2.8 times as many compared with corn, the smallest crop at 250 fields).
- Almost half of the data is dedicated to wheat (258) and corn (250), which are about the same size.
- Corn or wheat is second to soybean in both countries.

Field Distribution



- Argentina has 1.23 times as many fields as Brazil (665 vs 541), but they are spread over 69 blocks across a wide latitude range.
- Brazil's 541 fields are concentrated in 7 blocks inside a narrow area (about 1° of latitude and longitude). There are 199 fields in one block (Brazil_2).
- Soybeans are more percent of total consumption in Argentina (61.5% compared to 53.4%), and wheat is more percent of total consumption in Brazil (25.1% compared to 18.3%).
- Argentina's data is concentrated in 2020–2022 (400 of 665 fields). Brazil is strongest in 2017–2019 and 2023.

Seasonal and Country Imbalance



Imbalance Ratio

Comparison	Ratio
Fields of Argentina to Brazil	1.23
Largest and Smallest Crop	2.79
Blocks of Argentina to Brazil	9.86
Largest and Smallest Year(2021/2017)	2.92

Cross-validation fold balance

Fold	Argentina	Brazil	Total	Blocks
0	123	199	322	14
1	127	133	260	14
2	128	133	261	17
3	161	29	190	16
4	126	47	173	15
Total	665	541	1206	76

- 72% of all fields (863) are summer crops of soybean and corn, with the only winter crop being winter wheat. The season and crop are thus very well associated.
- Its second-crop corn is primarily Brazilian (73 of 85 fields) with only 12 Argentina fields.
- Wheat has no 2024 fields, and corn has only 17–20 fields per year in 2017–2019.
- The figures of results per fold vary, as 322 fields (199 Brazilian) are found in fold 0, but only 29 Brazilian fields are found in fold 3.
- The sample size of 1206 fields is too small for deep learning, and even smaller for each combination of crop, country and year.

5) Confusion Matrix Analysis

Model	SAR Macro-F1	Optical Macro-F1	Fusion Macro-F1	Improvement over Optical
Logistic Regression	0.847	0.862	0.948	+0.086
Random Forest	0.835	0.717	0.881	+0.164
SVM RBF	0.867	0.815	0.914	+0.099
XGBoost	0.848	0.790	0.903	+0.113

- All three classifiers' Macro F1 scores were improved with the Fusion model. Random Forest showed the highest increase of 0.164 followed by SVM and Logistic Regression.
- This improvement is due to the complementarity of the two satellite data sources. Optical features from Sentinel-2 provide information on spectral characteristics and the growth of crops while SAR features of Sentinel-1 offer information on crop structure and radar backscatter.
- The combination of both feature groups enables the classification models to account for a larger number of crop features. This helps lessen the constraint of depending only on one satellite data source and enhances the classification performance.

6) Statistical Significance Analysis using McNemar's Test

- McNemar's test was performed to compare the classification predictions of SAR-only, Optical-only and Fusion models using paired predictions from the same 1,206 fields.
- Out of 12 comparisons, 11 comparisons showed statistically significant differences ($p < 0.05$). The only non-significant comparison was between SAR and Optical Logistic Regression ($p = 0.1604$).

- Fusion showed statistically significant improvements over both the SAR-only and Optical-only approaches across all four classifiers.

7) Limitations

- Crop imbalance - 58% of fields is soybean and only 21% is corn and models may favour soybean and the minority fields need special attention.
- Country imbalance in spatial coverage: Argentina 69 blocks / Brazil 7 blocks. Brazilian data is localized around the small region that encompasses the fields (approximately 100 km wide across Brazil) and may not be representative of other Brazilian regions.
- Large blocks: 52% of the dataset is on the 5 biggest blocks, Brazil_2 has 199 fields in its large block. This is difficult to put into the context of balanced spatial folds (fold table).
- Temporal imbalance: yearly counts range from 78 (2017) to 228 (2021). Argentina has few fields in 2017–2019 and Brazil few in 2020–2022.
- Summer sowing of 72% of fields and second crop predominantly consisted of Brazilian corn (73 of 85), as well as a lack of wheat data for 2024.
- Season–crop confounding: almost all of the seasons are either pure season groups (winter = wheat only) or pure crop groups.
- Growing seasons vary across countries (average length: 168 days in Argentina, 146 days in Brazil, so a universal time frame is not applicable).
- Extraction of field-level SAR statistics were effected for empty/invalid field geometry.
- Coordinate interpretation: Coordinate components for 2D interpretation were required before building 2D geometry.
- Absence of Sentinel-1 acquisitions: Acquisitions were not available for some fields in the chosen time period.
- 10 missing NDRE values were found in two fields and they were handled using median imputation within the model pipelines.
- Model evaluation was performed using five-fold spatial cross-validation, and results may vary with different spatial partitions.

8) Conclusion and Next Week Plan

- The SAR-Optical Fusion approach got the best overall classification performance.
- Logistic Regression with Fusion achieved 95.77% accuracy and 94.79% Macro-F1 score.

- Fusion outperformed both SAR-only and Optical-only approaches across all four classifiers.
- The results showed that combining Sentinel-1 SAR and Sentinel-2 Optical features improves the classification of corn, soybean and wheat.

Next Week Plan

- Finalize the SAR-Optical feature fusion pipeline and evaluation.
- Review and interpret classification results and feature importance.
- Complete the final report with methodology, results and conclusions.
- Prepare the final project presentation.